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Adaptation in a variable environment: Phenotypic plasticity and bet-hedging during egg diapause and hatching in an annual killifish

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Two ways in which organisms adapt to variable environments are phenotypic plasticity and bet-hedging. Theory suggests that bet-hedging is expected to evolve in unpredictable environments for which reliable cues indicative of future conditions (or season length) are lacking. Alternatively, if reliable cues exist indicating future conditions, organisms will be under selection to produce the most appropriate phenotype —that is, adaptive phenotypic plasticity. Here, we experimentally test which of these modes of adaptation are at play in killifish that have evolved an annual life cycle. These fish persist in ephemeral pools that completely dry each season through the production of eggs that can remain in developmental arrest, or diapause, buried in the soil, until the following rainy season. Consistent with diversified bet-hedging (a risk spreading strategy), we demonstrate that the eggs of the annual killifish *Nothobranchius furzeri* exhibit variation at multiple levels—whether or not different stages of diapause are entered, for how long diapause is entered, and the timing of hatching—and this variation persists after controlling for both genetic and environmental sources of variation. However, we show that phenotypic plasticity is also present in that the proportion of eggs that enter diapause is influenced by environmental factors (temperature and light level) that vary seasonally. In nature there is typically a large parameter zone where environmental cues are somewhat correlated with seasonality, but not perfectly so, such that it may be advantageous to have a combination of both bet-hedging and plasticity.

KEY WORDS: **Coin-flipping, development, *Nothobranchius furzeri*, risk-spreading, seed bank, unpredictability.**

Different environmental conditions are likely to favor different modes of evolutionary response (Wong and Ackerly 2005; Simons 2011; Salinas and Munch 2012). Two modes of adaptation to variable environments are phenotypic plasticity and bet-hedging. A bet-hedging strategy maximizes (geometric) fitness multiplicatively across generations either through the production of a broad array of phenotypes (diversified bet-hedging) or a narrowly unimodal distribution of "safe" trait values (conservative bet-hedging) (Simons and Johnston 1997). Adaptive or predictive phenotypic plasticity is the phenomenon whereby a genotype produces different phenotypes depending upon environment (Ghalambor et al. 2007) thus allowing an organism to have higher fitness than if the trait value was fixed. Whether predictive phenotypic plasticity or bet-hedging is expected to evolve hinges critically upon the reliability of cues indicating future conditions (Wong and Ackerly 2005) and costs (Relyea 2002). Bet-hedging is expected to evolve in unpredictable environments for which reliable cues indicative of future conditions are lacking (Simons 2011). In other words, in an unpredictable world organisms are selected to "assume" the worst case scenario and evolve a "risk-spreading" or "safe" reaction norm or strategy, which can lead to reduced within-season fitness (relative to a nonbet-hedging strategy), but increased long-term geometric mean fitness (across multiple generations). Alternatively, if perfectly reliable cue(s) exist indicating future conditions, organisms will be under selection to produce the most appropriate phenotype so as to achieve higher

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fitness—that is, phenotypic plasticity (Cohen 1967; Wong and Ackerly 2005). However, natural environments may vary in their predictability (Colwell 1974) in a way that may select for a combination of bet-hedging and plasticity (Cohen 1967; Kaplan and Cooper 1984; Moran 1992; Wong and Ackerly 2005; Donaldson-Matasci et al. 2013). This combination of strategies has been explained in several equivalent ways, including "adaptive coin flipping" (Kaplan and Cooper 1984) and diversified bet-hedging around the norm of reaction (Wong and Ackerly 2005).

Many temperate zone insects produce diapause eggs in response to changing environmental cues indicating the onset of winter. Because the date of first frost will vary among years the insect must optimize the switch from producing direct developing eggs potentially capable of completing a second generation, to producing diapause eggs capable of surviving the winter (Cohen 1970; Taylor 1980). As the probability of winter's onset increases (decreasing daylight hours) so does the proportion of diapause eggs that are produced (Bradford and Roff 1993, 1997). When environmental cues are imperfect predictors of future environment, a graded shift from the production of direct developing to diapause eggs is expected. In other words, a measure of bet-hedging is expected in addition to phenotypic plasticity—ensuring that at least some offspring will survive (Cohen 1966; Kaplan and Cooper 1984; Moran 1992).

In addition to variability in whether diapause is entered, a staggered emergence from diapause (i.e., hatching or germination) could represent a bet-hedging strategy against unpredictability in season beginning date (Leon 1985; Venable 1989; Simons and Johnston 1997; Philippi et al. 2001). Simons and Johnston (1997) present the results of a simulation model set in the context of an annual plant found in a temperate region. The date of final spring frost varies among years. If a seed germinates early, and a frost occurs after it has germinated it will die. If a seed germinates very late, then it will avoid frost, but will have lower fecundity at the end of the season since it started growing late. When season beginning date was made more predictable the optimal germination strategy is to produce seeds that all germinate around the same date, right after the final frost. In contrast, when date of final frost is highly unpredictable between years a bet-hedging strategy is favored where seeds germinate over a broad distribution of time. This way no matter what time the season starts there are some seeds able to capitalize.

There are other seasonal patterns that organisms have adapted to besides winter, especially in the large portion of the world where winter freezing does not occur. Killifish adapted to life in ephemeral aquatic habitats are found across large portions of Africa and South America in habitats ranging from open savannah to forest (Murphy and Collier 1997; Hrbek and Larson 1999). Due to the regular or periodic drying of their aquatic habitat, these fish have evolved eggs capable of undergoing prolonged diapause or reversible developmental arrest at specific stages during development (Peters 1963; Wourms 1972a, b, c). As their aquatic environment dries out and the fish die, the developing eggs remain viable, buried in the soil, until the following rainy season (Genade et al. 2005; Watters 2009). In habitats that have distinct wet and dry seasons, where aquatic pools regularly dry each year and hatching, maturation, and reproduction all occur within a single season, this is an annual life cycle. Annual killifish face analogous selection pressures as the annual plants and insects described above. Specifically, the start time and total length of the rainy season shows variation both within and across seasons (Watters 2009). Within a season, rain may come in pulses such that pools that support fish have multiple inundations and dryings (Watters 2009; Polacik et al. 2011; Polacik et al. 2014). Killifish inhabiting such environments face the risk of false starts in regards to season beginning—an early rain followed by a period of dryness could cause eggs to hatch only to have the fish perish quickly thereafter. Between seasons the duration of the rainy season and total annual rainfall can also vary (Podrabsky et al. 1998; Watters 2009; Polacik et al. 2011). Given favorable conditions it may be possible for direct-developing embryos to complete a second generation during a single season. At the other end of the spectrum, it is possible that a given year may occasionally have such low or sporadic rainfall that aquatic pools occur too briefly for fish to successfully reproduce (Pienaar 1968; Wourms 1972c; Podrabsky et al. 1998; Watters 2009).

The annual killifish studied here, *Nothobranchius furzeri*, is native to savannah habitat of southern Mozambique and Zimbabwe where it inhabits temporary aquatic pools which dry on an annual basis (Fig. 1). *Nothobranchius furzeri* eggs are capable of undergoing diapause, developmental arrest with an accompanying slowing of metabolic rate (Podrabsky et al. 2010b), at three specific stages during embryology—termed diapause I, II, and III (depicted in Figs. 2 and 3). Diapause I occurs early in development and is brought about by embryos being exposed to harsh environmental conditions, including cold temperatures, hypoxia, and presence of adult fish (Wourms 1972c; Inglima et al. 1981; Levels et al. 1986; Podrabsky et al. 2010b). Diapause II can occur in embryos possessing 38 somite pairs, just after a rudimentary tubular heart and the beginnings of several other organ systems have formed. It is during this diapause that embryos are most resistant to temperature extremes, desiccation, and oxygen deprivation (Matias and Markofsky 1978; Matias 1982; Podrabsky and Hand 1999). Diapause III can occur when the embryo is fully developed and capable of hatching. The embryos of *N. furzeri*, and other annual killifish, are capable of either entering into or skipping each phase of developmental arrest giving rise to a diversity of different developmental pathways (Fig. 3; Wourms 1972c). We refer to embryos that enter into diapause II as following the diapause pathway and those that skip this state and instead

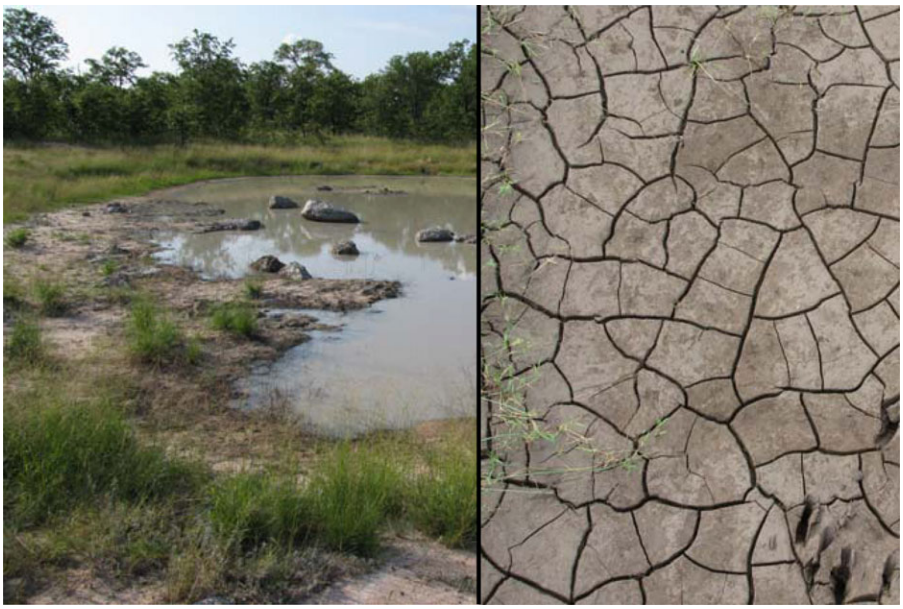


Figure 1. *Nothobranchius furzeri* habitat in the wet (left panel) and dry (right panel) seasons.



Figure 2. *Nothobranchius furzeri* male and female. Embryos that have arrested development in diapause I, II, and III.

proceed directly to diapause III as following the direct-developing pathway (Podrabsky et al. 2010a).

Here, we carry out a series of experiments on egg diapause induction and hatching variability designed to test two modes of adaptation to environmental unpredictability, phenotypic plasticity and bet-hedging. We are interested in whether the induction of diapause is sensitive to environmental cues that may indicate timing in the year (phenotypic plasticity), or if it shows intrinsic variability as a strategy to minimize risk of complete reproductive failure in an environment where timing and length of the rainy season can vary by years (bet-hedging). If a bet-hedging strategy is being employed we would predict variability in developmental pathway and hatching date. The following specific bet-hedging predictions were tested: (1) eggs from the same male–female pair, spawned and fertilized on the same day and treated in an identical manner, are capable of undergoing heterogonous development rates (i.e., some enter diapause and some exhibit direct-development), (2) not all eggs hatch during the first rainfall event (i.e., when wetted the first time) even if development is complete, and (3) eggs that do not hatch remain viable for long periods, and hatch under identical circumstances at a later date.

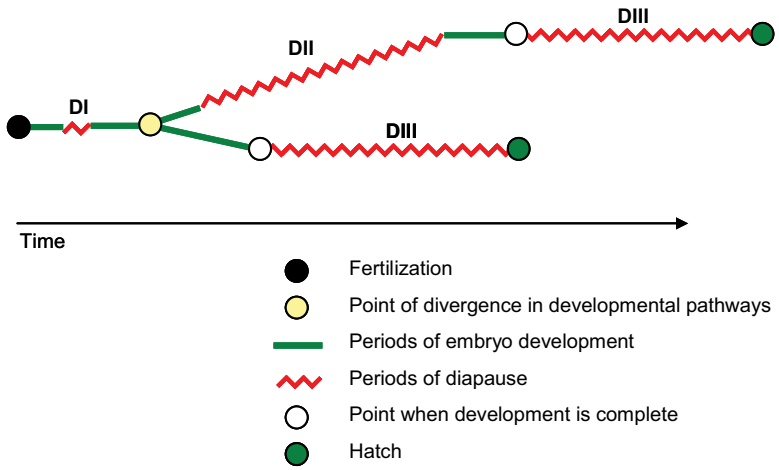


Figure 3. **Alternative developmental pathways for annual killifish embryos (adapted from Wourms 1972c; Watters 2009). Several days after fertilization all embryos pass through a dispersed cell phase in which development can be arrested (Diapause I). Environmental conditions including hypoxia or cold temperatures have been shown to induce an arrest in or significant lengthening of this phase. However, given incubation conditions in the present study, the dispersed cell phase was rather transient, generally lasting only a few days. After cellular reaggregation and the formation of the somite-embryo, a point of divergence is reached (yellow circle) where embryos become committed to following either the diapause II pathway (upper trajectory) or direct-developing pathway (lower trajectory). This point is depicted as a bifurcating fork because here embryos following the different pathways begin to diverge morphologically and physiologically. Direct-developing embryos proceed with development until it is complete and they are capable of hatching given appropriate environmental conditions (lower white circle). Embryos following the diapause II pathway arrest development when they attain approximately 38 somite pairs. After a variable length of time, diapause II embryos "escape," resume development, and reach the stage where they are capable of hatching (upper white circle). Fully developed embryos (white circles) are capable of hatching, yet can remain in this "ready and waiting" state (Diapause III) for variable lengths of time.**

If phenotypic plasticity were occurring we would expect that whether or not eggs enter diapause would be sensitive to environmental cues that indicate timing in the season. Our goal was to select several ecologically relevant environmental cues and expose eggs to combinations of these cues that mimic realistic levels at different times in the season. We obtained long-term monthly precipitation, temperature, and day length data from a weather station 94 kilometers from our *N. furzeri* collection locality (Table S1). These data (Fig. S1) allowed us to parameterize our cues, or experimental treatments to encompass the range of environmental variability that eggs of *N. furzeri* would ordinarily experience in their natural habitat over the course of a year. The dry season is characterized by cool temperatures and reduced day length while the rainy season is associated with warm temperatures and longer days (Fig. S1, Table S2). We therefore anticipated that cooler temperatures and shorter day length will cause embryos to enter diapause and warmer temperatures and longer day length will be associated with direct development.

Methods

PHENOTYPIC PLASTICITY IN RESPONSE TO ENVIRONMENTAL CUES

The *N. furzeri* population used in these laboratory studies were originally collected from an ephemeral pool in the Chefu river drainage in southern Mozambique near the border with Zimbabwe. Mature F2 generation male and female *N. furzeri* were maintained in 40 gallon stock tanks in a fish room (25 degrees C, 12 hours light: 12 hours dark daily cycle) and provided with spawning substrate. Eggs were collected daily, over a one month period, and distributed individually into the wells of 48-well tissue culture plates containing Yamamoto's embryo incubation medium (Valenzano et al. 2009). Eggs were incubated at three temperatures (20, 25, and 30°C) crossed with four light treatments (hours light: hours dark—0:24, 10:14, 12:12, 14:10) for a total of 12 different combinations. Our 12 treatment combinations of temperature and light were replicated for a total of 24 environmental chambers that each contained one tray with 48 embryos. Since fish embryos are translucent it is possible to stage them throughout development (Wourms 1972a). Eggs were observed once or twice weekly under a dissecting microscope and scored for if, when, and what stage they entered diapause. Diapause II was apparent as an arrest of development at the 38 somite stage. Direct-developing embryos bypass this arrest and instead continue developing until diapause III is reached. Embryos following the diapause and direct-developing pathway exhibit significant morphological divergence that begins prior to the stage at which embryos enter into diapause II (Podrabsky et al. 2010a; Furness et al. 2015). The date at which diapause II embryos resumed development or "escaped" from diapause was recorded. Lastly, we recorded the

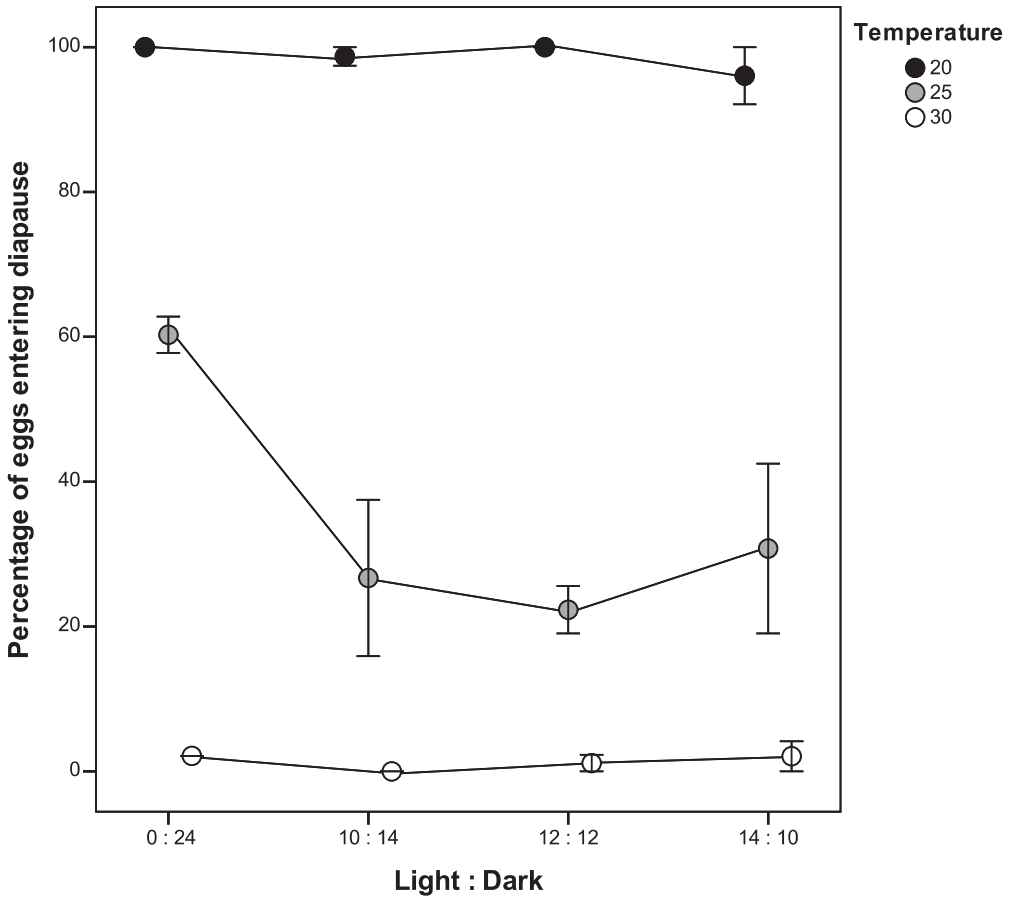


Figure 4. Percentage of *N. furzeri* embryos that entered diapause II (± SE of tray means) as a function of incubation conditions (temperature and day length). Total embryos assayed was $n = 1152$ of which 127 went bad prior to determination of developmental pathway resulting in a sample size of $n = 1025$.

date at which eggs either hatched or were no longer viable. These data provide a longitudinal record of development for embryos incubated at different temperature and light levels.

We analyzed the data in two complementary ways. For each of the dependent variables (percentage of embryos that entered diapause II, days until "escape" from diapause II, and total development time) we calculated tray means by averaging values for the 48 embryos contained within each tray. We then used general linear models with temperature, light, and the interaction as fixed effects. This procedure homogenizes variance between groups, but reduces power because each tray becomes the unit of analysis rather than the 48 individual eggs contained within. Alternatively, we used generalized linear models with temperature, light, and their interaction as fixed effects and individual embryos nested within tray. For the analysis of how temperature and light affect developmental pathway (0 = diapause, 1 = direct development) we used a binomial distribution and logit link function, while for the continuous variables (days until "escape" from Diapause II, and total development time) we employed a gamma distribution and log link function.

BET-HEDGING DURING DIAPAUSE INDUCTION AND HATCHING

Diversified bet-hedging requires variation in phenotype when similar genotypes are exposed to the same environment (Philippi and Seger 1989). Thus, for this experiment we kept track of parentage. We incubated cohorts of eggs of known parentage in a common garden environment and examined phenotypic variance in developmental pathway and timing of hatching.

Five male–female pairings were generated and these fish were reared in separate tanks containing spawning substrate. Eggs were collected daily from each tank and distributed individually into the wells of a 48-well plate containing compacted, moist peat moss. These trays were incubated in a dark cupboard at 25 degrees C. Approximately twice weekly embryos were examined such that developmental pathway—direct developing or diapause—could be ascertained. When eggs first reached Diapause III—indicated by fully developed eyes lined with a gold iris (Fig. 2) we attempted to induce hatching by placing them in dishes filled with aquarium water and peat moss extract. Eggs that did not hatch in the 24-hour period were returned to their respective

peat-containing well. We repeated identical hatching attempts weekly thereafter. This iterative pattern was repeated until all eggs had hatched. These hatching trials were designed to simulate what is likely to occur at the beginning of the rainy season—occasional short bursts of rain that briefly inundate embryos followed by a return to drier conditions (incubation on moistened peat moss substrate).

Results

PHENOTYPIC PLASTICITY

Diapause I (i.e., the dispersed cell phase) was transient under the conditions of our experiment. It generally lasted only a few days, sometimes a bit longer in the 20°C treatments. Both temperature and light influenced whether eggs entered diapause II or were direct developing (GLM analysis, temperature $F_{2,12} = 399.3$, $P < 0.0001$, light $F_{3,12} = 4.6$, $P < 0.05$). At 20 C (a temperature that eggs typically experience during the dry season—see Fig. S1), virtually all eggs entered into diapause II regardless of light level (Fig. 4). In contrast, at 30 C (a temperature that would likely only be found in the middle of the wet season) virtually all eggs were direct developing, again regardless of light level. At the temperature of 25°C, a variable proportion of embryos entered diapause while the others were direct developing (Fig. 4). It was only at 25 C that light level affected the proportion of embryos that entered diapause—causing the significant temperature × light interaction ($F_{6,12} = 3.8$, $P < 0.05$). Specifically, in the dark treatment 60.3% of embryos entered diapause II, while at the three light treatments (10, 12, and 14 hours) 26.7, 22.3, and 30.8% of embryos entered diapause, respectively. Thus, temperature was the sole determinant of developmental pathway at the lowest and highest temperatures. It was only at the intermediate temperature that an effect of light level was observed.

We limited the dataset to embryos that entered diapause and asked whether light and temperature had an effect on the number of days until such embryos "escaped" from diapause II. The mean number of days until "escape" across all treatments was 126.7 [range: 44–639 days]. Temperature, light level, and the temperature x light level interaction were all significant in regards to length of diapause (Generalized LM analysis, all $P < 0.01$). Although temperature had a dominant influence on whether diapause was entered, it had a lesser influence on its length (Table S4). This result should however be interpreted with caution because few embryos entered diapause at 30°C (Fig. 4) and because the temperature of the 20°C environmental chambers increased to 25°C (ambient room temperature) for a 72-hour period approximately three months into the experiment which likely caused some embryos to prematurely escape from diapause II. Light level had a significant impact on the length of diapause II with embryos in the dark showing significantly longer periods of arrest. This trend was particularly apparent at the intermediate temperature of 25°C (Table S4).

Total development time is the number of days from fertilization until embryos hatch or perish. *Nothobranchius furzeri* embryos that enter diapause II have significantly longer development periods relative to embryos that follow the direct-developing pathway (Fig. 5)—such a result is found whether the analysis is performed on the subset of embryos that hatched, those that perished, or both categories of embryos. We then asked how temperature, light level, and developmental pathway affected long-term embryo survival. For these survival analyses we excluded embryos that perished before developmental trajectory could be determined and embryos that hatched during the course of incubation. Results are qualitatively the same regardless of whether such groups of embryos were included. Embryos that entered diapause II ($n = 321$) survived significantly longer than direct developing ($n = 387$) embryos (Kaplan–Meier method, Mantel–Cox Test, $\chi^2 = 331.3$, $P < 0.00001$). Embryo survival distributions at the three different temperatures were also significantly different (Kaplan–Meier method, Mantel–Cox Test, $\chi^2 = 222.8$, $P < 0.00001$). Embryos incubated at 20 degrees ($n = 218$) had a mean survival time of 161.1 (5.6 SE) days, those at 25 degrees ($n = 230$) had mean survival of 137.9 (SE 9.3) days, and those at 30 degrees ($n = 260$) 74.3 (SE 2.6) days. These significant differences are largely due to the effect temperature had on the proportion of embryos that entered diapause II (i.e., nearly all did so at the lowest temperature and very few at the warmest; Fig. 4) and the significantly longer survival of embryos that entered diapause II (Fig. 5).

BET-HEDGING

Nothobranchius furzeri eggs from the same male female pair and clutch, incubated under identical common garden laboratory conditions (25°C, dark, on peat moss) are capable of undergoing heterogeneous development rates (i.e., some enter diapause II and some follow direct-developing trajectory)—Figure 6A. Furthermore, eggs that enter into diapause II remain in this state for variable lengths of time, so the emergence from diapause II is not synchronous. Once embryos reached diapause III not all eggs hatched at the same rate, even though they were at a stage where development was complete and thus presumably physically capable of hatching (Fig. 6B). Lastly, eggs that did not hatch initially remained viable and hatched under identical conditions experienced at a later date (Fig. 7). There were significant differences among the five male–female pairs in both the proportion of diapause versus direct-developing embryos (generalized linear model, binomial distribution, and logit link function, Wald $\chi^2 = 23.0$, $P < 0.001$) and the number of consecutive once-weekly inundations (hatching attempts) before successful hatching (generalized linear model, Poisson distribution and log link function, Wald $\chi^2 = 59.3$, $P < 0.001$).

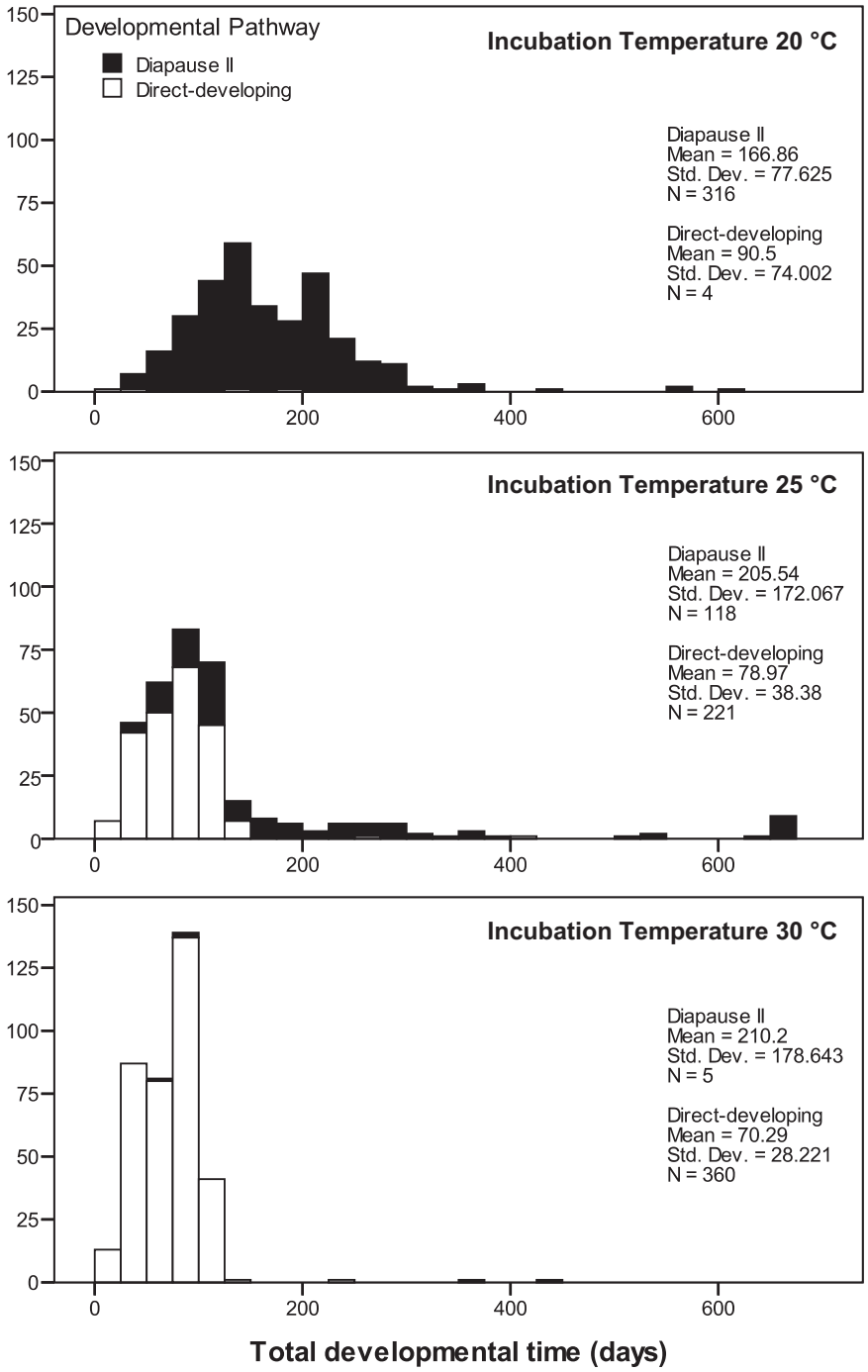


Figure 5. **Stacked frequency histograms of total development time of *N. furzeri* embryos incubated at 20, 25, and 30 degrees Celsius. Total development time is defined as number of days from fertilization until embryo hatched or perished. Embryos that entered diapause II are indicated in black and those that were direct-developing are indicated in white. Within each temperature treatment all photoperiods have been pooled together.**

Discussion

VARIATIONS ON THE THEME OF ANNUAL LIFE CYCLES, ENVIRONMENTAL VARIATION, AND THEORETICAL CONTEXT

The life cycle of killifish adapted to life in ephemeral water bodies is more typically associated with annual plants (Philippi 1993; Clauss and Venable 2000; Venable 2007), invertebrates found in ephemeral aquatic habitats (Hairston and Caceres 1996; Simovich and Hathaway 1997) or insects found in temperate regions with freezing winters (Bradford and Roff 1997; Hopper 1999; Soula and Menu 2005; Fielding 2006). In annual killifish many of the same questions can be addressed as in these analogous study

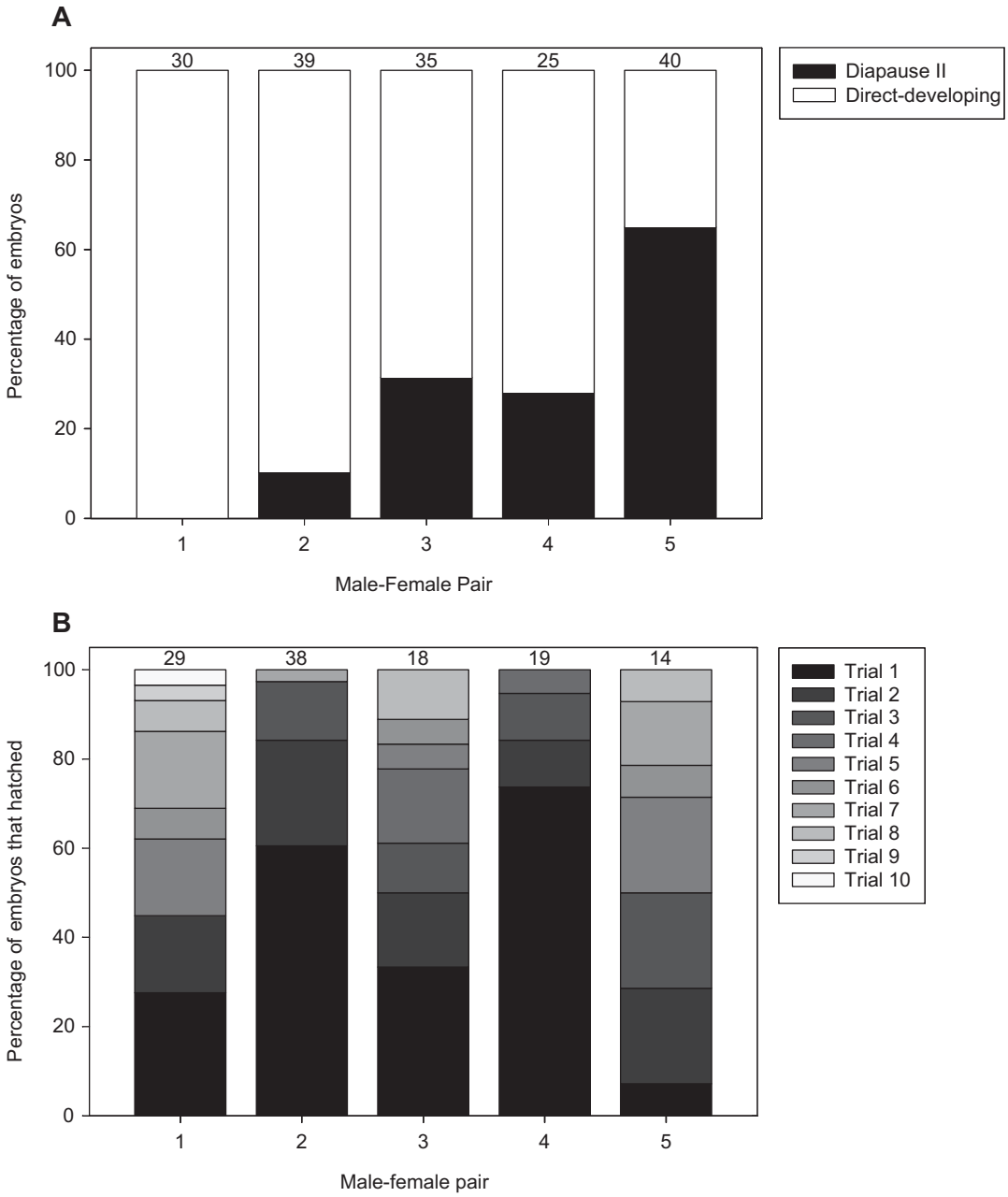


Figure 6. **(A) *Nothobranchius furzeri* embryos from the same male–female pair (1–5), incubated under identical conditions (25°C, dark, peat moss) exhibit diversity in developmental pathway. Dark bars represent diapause II embryos and white bars direct-developing embryos. Numbers above bars are number of embryos assayed for each pair. (B) After embryos reached the prehatching stage (DIII) they underwent once-weekly hatching trials in which they were inundated with water for a 24-hour period. Embryos exhibited variation in the number of consecutive trials before hatching.**

systems with the added benefit that one can observe development through their translucent eggs. Here, we demonstrate that multiple stages of embryonic diapause yield remarkable flexibility in the duration of development with *N. furzeri* embryos exhibiting a combination of both phenotypic plasticity and bet-hedging strategies.

Annual killifish embryos face a number of decisions including: whether to enter into early and midembryonic diapause (I and II), how long to remain in this state of developmental arrest, and once development is complete (diapause III), when to hatch. These decisions must be made in the context of aquatic pools that vary in the timing of filling and their duration. Like other organisms that reside in annual environments, the habitat in which *N. furzeri* resides is characterized by both within and between season environmental variability. All pools will be dry for a prolonged period during the dry season, but may also fill and

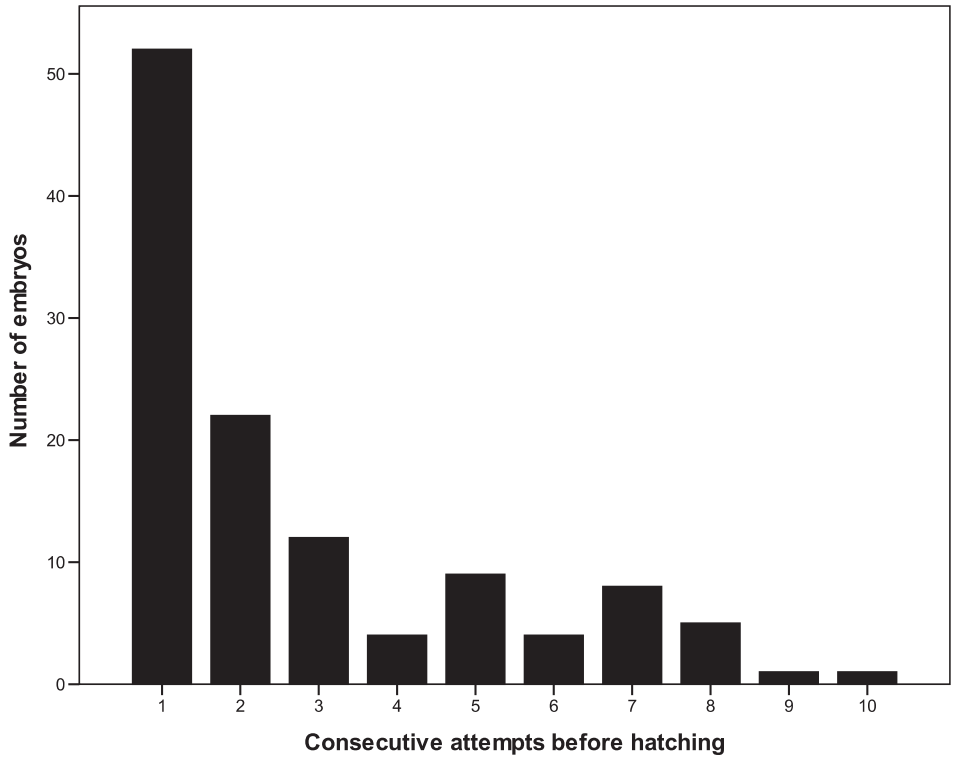


Figure 7. Hatching distribution for *N. furzeri* embryos (n = 118) that reached the pre-hatching stage (Diapause III). Although fully developed and presumably physically capable of hatching such embryos exhibit variation in the number of consecutive (once weekly) wetting trials or attempts before hatching.

dry repeatedly during the wet season (Polacik et al. 2014). Consideration of these multiple levels of variability has led to three interrelated questions and modeling efforts that are summarized in Table 1. What is the optimal annual germination / hatching fraction given probability of good versus poor years (Cohen 1966; Venable 1989)? When and in response to what cues should there be a switch from direct developing to diapause eggs given heterogeneity in length of growing season, and what is the nature of this transition, graded or bang-bang (Cohen 1970; Bradford and Roff 1997; Spencer et al. 2001; Halkett et al. 2004)? Lastly, what is the optimal within year germination/hatching strategy given risk of season false-starts, that is what percentage of eggs are expected to hatch at each rainfall event (Leon 1985; Venable 1989; Simons and Johnston 1997; Philippi et al. 2001)? An important consideration pertaining to each of these questions is how cue reliability (within and between generations) affects the evolved balance between adaptive (predictive) plasticity and bet-hedging (Wong and Ackerly 2005; Donaldson-Matasci et al. 2013).

EMBRYO PHENOTYPIC PLASTICITY

The first decision a newly fertilized embryo faces is whether to enter into or skip early and mid-embryonic diapause (I and II). This decision will have substantial fitness implications via its effect on long-term survival prospects and embryos would be expected to have evolved the capacity to perceive and use the most reliable environmental cues to best inform their decision (Cohen 1967; Donaldson-Matasci et al. 2010). Specifically, Diapause I and II may be specialized for allowing embryos to traverse the rather long gap between rainy seasons, which is likely to last at least six months for *N. furzeri* (Fig. S1, Table S1). In contrast, Diapause III may represent a means by which a fully developed embryo can, over a comparatively shorter time-frame, remain "ready and waiting" for circumstances in which to hatch.

Whether eggs of *N. furzeri* enter into diapause II or are direct-developing exhibits plasticity to temperature, and at intermediate temperature, to day length. It is the historical association between environmental cue(s) and timing of year which natural selection would be expected to act upon in determining the optimized reaction norm for entering into or skipping diapause. For temperate zone insects, photoperiod is evidently a reliable index of timing in the year in relation to winter (Adkisson 1964; Saunders 1987) and may thus preclude the use of other cues. *Nothobranchius furzeri* embryos instead appear more responsive to temperature, perhaps because the variation in photoperiod at this latitude is limited (Fig. S1) relative to a temperate region. This predominance of temperature as a cue offers a potential explanation for the fact that very warm or very cool incubation temperatures induce nearly

Table 1. Summary of prominent previous modeling efforts regarding bet-hedging and phenotypic plasticity associated with diapause; application to the annual killifish study system is indicated, general predictions are derived, and evidence summarized.

General question	What is optimal interannual hatching strategy given probability of good versus poor year?	What is optimal within-season hatching strategy given variation in season beginning date and risk of false starts?	At what time of year or given what environmental cue(s) should embryos switch from direct developing to entering diapause given heterogeneity in growing season length?
Variables in the model	(1) Probability of good versus poor year. (2) Fecundity of embryos that hatch in good versus poor years. (3) Embryo survival rate in soil bank.	(1) Probability of season false-start (pool fills but quickly dries before hatched embryos capable of attaining maturity) or variance in season beginning date. (2) Fitness of embryos that hatch at different times. (3) Short-term embryo survival rate in soil bank.	(1) First generation to hatch in wet season produces embryos capable of entering diapause (and surviving to next season) or skipping diapause (and having potential to complete second generation during same season). (2) Only embryos in diapause (originating from both potential generations) can survive dry season. (3) Season length varies between years.
Killifish assumption(s)	(1) Variation in year quality. (2) In poor years ephemeral pools are of such short duration that hatched embryos are unable to reach reproductive maturity prior to pool dry down, hence no embryos are added to egg bank and fecundity is zero for all hatched embryos. In good years fecundity is relatively high; annual killifish exhibit near continuous daily reproduction once maturity is reached with fecundity increasing as a function of body size. (3) Unhatched embryos in the soil bank in Diapause I and II have moderate to high long-term survival.	(1) Within-season variance in rainy season beginning date and hence 'true' pool filling date. (2) Embryos that hatch during false pool filling (i.e., one that doesn't last longer than 20 days) have fitness of zero. (3) Embryos that have completed development (Diapause III) but do not immediately hatch have high short-term survival.	(1) Embryos can enter into or skip diapause. (2) Entry into diapause is required to traverse the dry season. (3) There is between-year heterogeneity in the duration of ephemeral aquatic pools and whether or not there are multiple pool fillings within a single wet season.
General killifish prediction[1]	Most, but not all, embryos will hatch each year (i.e., a multi-year soil egg bank is expected).	Not all embryos will hatch each rainfall event even if development is complete.	Skipping diapause (i.e., direct-developing pathway) only has the potential to be beneficial early in the rainy season when temperatures are high and day length long. Toward end of the rainy season all embryos should enter diapause.
Experimental evidence	Embryos capable of surviving for longer than two dry seasons in laboratory environment (Fig. 5); Other literature sources indicate even longer survival times possible in laboratory (Simpson 1979; Watters 2009), and field observations that are consistent with multi-year embryo bank have been reported (Pienaar 1968; Wourms 1972c; Simpson 1979; Watters 2009)	Embryos that attained hatching capability exhibited high degree of variance in hatching time (Figs. 6, 7)	In laboratory environment embryos exhibit alternative developmental pathways (direct developing or entering into diapause). Embryos more likely to skip diapause given environmental cues indicative of the wet season (Fig. 4). In contrast embryos given cues likely to only be found in the dry season (low temperatures and short days) all entered diapause. See also Podrabsky et al. (2010a,b).

(Continued)

Table 2. Continued.

General question	What is optimal interannual hatching strategy given probability of good versus poor year?	What is optimal within-season hatching strategy given variation in season beginning date and risk of false starts?	At what time of year or given what environmental cue(s) should embryos switch from direct developing to entering diapause given heterogeneity in growing season length?
Specific predictions[2]	Within population: Could predict the percentage of embryos expected to hatch each year from a multi-year egg bank. Specifically, given several simplifying but reasonable assumptions, the proportion of embryos that hatch each year is predicted to equal the long-term probability of it being a good year (Cohen 1966). Comparative: Populations from more arid regions, where the probability of a poor year is higher, will have larger/more prominent multi-year egg bank relative to populations from more humid regions.	Within population: Given quality pool duration data, could predict the percentage of embryos expected to hatch at each wetting event. Comparative: The more unpredictable the season beginning date (i.e. greater risk of false starts) the more variance in hatching time is expected (i.e. lower percentage of embryos predicted to hatch at each rainfall event / pool filling)	Within population: Could theoretically predict the mean switching date and nature of transition (gradual or abrupt) from direct developing to diapause. Other environmental cues indicative of approaching dry season will tend to induce diapause. Comparative: Diapause will be more prominent/readily induced in populations from arid regions with strong seasonality. In contrast, direct development will be more prominent/readily induced in populations from humid regions with mild seasonality.
Theoretical underpinning	Cohen (1966); Cohen (1967); Brown and Venable (1986); Clauss and Venable (2000); Philippi et al. (2001).	Cohen (1966); Leon (1985); Venable (1989); Simons and Johnston (1997).	Cohen (1970); McNamara (1994); Bradford and Roff (1997); Spencer et al. (2001).

[1]Based on coarse parameter estimates summarized in Tables S2 and S3.

[2]We currently lack detailed data that connects rainfall patterns to the timing of filling and duration of ephemeral aquatic pools. These data are need to appropriately parameterize a quantitative model. Here, we describe the specific type of predictions that could be tested within populations or comparatively among populations that differ in seasonality.

all eggs to follow a single developmental trajectory; these strong cues may be associated with relative certainty of timing in the year. Persistent temperatures of 30°C are only predictably associated with the wet season and cause all embryos to skip diapause II (direct development). Persistent temperatures of 20°C are only predictably associated with the dry season and cause all embryos to enter diapuse II. Photoperiod is only a factor that influenced the developmental fate of eggs at the intermediate temperature of 25°C, but even at this temperature this effect was only apparent in the dark treatment (Fig. 4).

Previous work in other annual killifish has demonstrated that a variety of environmental factors including temperature (Levels and Denuce 1988; Podrabsky et al. 2010a), photoperiod (Levels and Denuce 1988; Podrabsky and Hand 1999), and whether embryos are incubated in tanks containing adult fish (Inglima et al. 1981) can influence the probability of embryos entering or skipping diapause (reviewed in Podrabsky et al. 2010b). Currently, we lack information on the interaction between maternal effects and offspring (embryo) plasticity but it appears that environmental conditions experienced by parents such as photoperiod (Markofsky et al. 1979) or maternal age (Podrabsky et al. 2010a) can modify embryo reaction norms—termed transgenerational plasticity (Salinas and Munch 2012). In the present study, individual male–female pairs "produced" different proportions of diapause versus direct-developing eggs (Fig. 6) when embryos were reared in a common garden environment at the intermediate temperature of 25°C. Since environmental effects were standardized (for both parents and embryos) this interindividual variation indicates that parentage (genetics or parental effects) influences the propensity for eggs to enter diapause. However, strong environmental cues experienced by the embryo can still override these parental effects (Fig. 4; Levels and Denuce 1988). Why is this the case? The critical issue may be the strength of autocorrelation between environmental cues and seasonality, how easily these cues are perceived by parents and embryos, and the constancy of the relationship between environmental cues and seasonality over evolutionary time (Wong and Ackerly 2005). A mother "deciding" whether to produce diapause or direct developing eggs will be relying on different, and more outdated, information than an embryo buried in the soil. Thus it may be advantageous for individual embryos to use phenotypic plasticity to adjust developmental trajectory (phenotype).

Whether the relationship between environment and offspring phenotype is framed from the perspective of the mother or the individual offspring can dictate interpretation and preferred terminology. Our comparison between bet-hedging and phenotypic plasticity naturally leads to the questions: Who's hedging their bets? Whose phenotype is plastic? A mother can have offspring with different phenotypes while using reliable cues to bias her "bets" in an adaptive manner. The use of the term diversified bet-hedging seems most intuitive when referring to this cohort-level variation, because it is in this context that we can easily see multiple bets being made (in the form of offspring expressing alternative phenotypes). Alternatively the individual embryo only gets to bet once (i.e. expresses a single phenotype) and is expected to use environmental cues to produce the most appropriate phenotype (plasticity). However, if cues are uninformative or unreliable the best an embryo can do is express one or the other phenotype with equal probability (i.e., flip a coin). This meshing of strategies encapsulates a measure of both predictive plasticity and inherent stochasticity and has been called coin-flipping plasticity/adaptive coin-flipping when viewed from the individual perspective but termed diversified bet-hedging around the norm of reaction when viewed from the parental perspective. We view these sets of terminologies as expressing the same general sentiment but reflecting a difference in perspective.

Given the relatively short-duration environments these fish occupy, why do embryos follow the direct-developing pathway? Why don't all embryos enter into each diapause as a conservative strategy? One possibility is that in nature the direct-developing pathway is a rarer occurrence than observed in a laboratory setting. Embryos buried in the mud at the bottom of a pool could experience several cues simultaneously including low oxygen levels, moisture levels, temperature, light, and the chemical cues of adult fish (Watters 2009; Podrabsky et al. 2010b). This may cause embryos to more readily enter into each successive diapause phase (I, II, and III) than in a laboratory environment where embryos are collected at regular intervals and incubated under relatively benign conditions.

A second nonmutually exclusive possibility is that it is adaptive for *N. furzeri* to have (some) direct-developing eggs early in the season and eggs that enter into diapause late in the season. Temperate zone insects are known to switch mid summer, from producing direct-developing eggs capable of completing a second generation to producing diapause eggs capable of surviving the freezing winter (Cohen 1970; Taylor 1980; Bradford and Roff 1993). Likewise offspring that hatch from nondiapausing eggs early in the rainy season may be capable of reproducing before the pools dry, causing a multiplicative increase in the fitness of the parent. Or, as is perhaps more likely to be the case given the short and variable aquatic environments of *N. furzeri*, pools may experience several inundations and dryings within a single rain season (Polacik et al. 2011; Polacik et al. 2014). Direct-developing embryos would be in the best position to capitalize on such a scenario since they reach hatching capability much more quickly than those that enter Diapause II.

The likelihood of *N. furzeri* being physically capable of completing a second generation within a single rain season or capitalizing on a second pool filling is feasible given that this species has an incredibly short-generation time and condensed life cycle (Terzibasi et al. 2007). This species can reach sexual maturity in as little as 3 weeks (Blazek et al. 2013). At 30°C, it took direct-developing embryos approximately 15 days from fertilization until development was complete and hatching capability was reached. Thus a minimum time for two generations to be completed in a single rainy season would be around 60 days (21 days for first generation to reach maturity assuming they hatch immediately upon first pool filling + 15 days embryo development time + 21 days for second generation to attain maturity; Table S3). Tozzini et al. (2013) recorded a mean pool duration of 73 days in one of the more arid regions in which *N. furzeri* is native and a mean of 142 days in a more humid region. Furthermore, Polacik et al. (2014) documented a case in which a pool initially remained filled for longer than one month, fully dried out, and refilled for a comparable period, all within a single rainy season; these authors further showed that *N. furzeri* hatched from embryos after a 50 day embryonic development exhibit an accelerated life-history characterized by faster growth, earlier maturity, and more rapid aging relative to fish hatched from embryos that had a 200 day embryonic development.

DIVERSIFIED BET-HEDGING OR THE "COIN-FLIPPING" COMPONENT OF PLASTICITY

In contrast to predictive phenotypic plasticity, diversified bet-hedging or risk-spreading strategies are expected to evolve given unreliable information. Environmental cues are sometimes imperfect indicators of future environment or season length, such that a diversified bet-hedging strategy can evolve on top of plasticity as a response to the inherent uncertainty (Kaplan and Cooper 1984; Moran 1992; Wong and Ackerly 2005; Donaldson-Matasci et al. 2010, 2013). Our argument for bet-hedging is twofold. At the intermediate temperature of 25 C (which may correspond to uninformative information regarding timing in the year) there was variation in whether Diapause II was entered or skipped. Secondly, once development is complete (DIII), there is additional variation in how long it takes for embryos to hatch. Here, embryos face the risk of season false starts in which pools fill but dry quickly thereafter such that any hatched embryos perish prior to reaching maturity. As with the decision to enter or skip Diapause II, embryos are predicted to be under selection to use any available cues indicating whether a particular pool filling will be long lasting and hence favorable for hatching (Cohen 1967; Leon 1985; Venable

1989; Clauss and Venable 2000; Donohue et al. 2005; Tielborger and Valleriani 2005). Given our experimental design we observed a high degree of variance in hatching time (Fig. 7). Both types of variation (whether or not to enter Diapause II, and hatching time once attaining Diapause III) were observed within the clutches of individual females when embryos were incubated under identical common garden conditions (Fig. 6) and is thus consistent with a diversified bet-hedging or coin-flipping response.

Simons (2011) created strength of evidence categories for demonstrating within-population bet-hedging. His criteria included: (1) identifying a candidate bet-hedging trait, (2) observing an unpredictable environmental factor, (3) demonstrating expression of bet-hedging trait at the genotype-level, (4) establishing variable fitness consequences, (5) demonstrating such consequences are advantageous under fluctuating selection, and finally (6) finding a quantitative match between bet-hedging trait expression and degree of fluctuating selection. We fulfill the first three of these criteria. We identified traits (developmental pathway and hatching time) that show a high degree of variance. We demonstrated an environmental factor, duration of aquatic environment that leads to a measure of habitat unpredictability (Fig. S1, Table S2). Lastly, we demonstrated variation in developmental pathway and hatching time is expressed within the same cohort (while controlling or minimizing genetic and environmental sources of variation). This latter point is important because variation expressed among individuals at the population-level (rather than the genotype) could result from genetic polymorphism—which is not considered bet hedging (Seger and Brockmann 1987; Philippi and Seger 1989; Simons 2011). In sum, we demonstrated that eggs with the same parentage (genetics), incubated under identical common garden laboratory environment (same environment) are capable of following either the diapause or direct-developing pathways and exhibit variation in hatching time. In the context of the natural history of these fish such variation in development and hatching time is likely an adaptation to environmental unpredictability (see Table 1) yet more rigorous quantitative tests await further gathering of appropriate environmental data and modeling effort. In Table 1, we detail the type of specific quantitative predictions that could be generated and empirically tested either within populations or comparatively across populations that differ in environmental variation (see also Clauss and Venable 2000; Rajon et al. 2014).

Desert annual plants that experience considerable variation in year quality often exhibit multi-year seed banks, in which not all seeds germinate each year (Philippi 1993; Venable 2007). It is conceivable that similar complexity exists in the embryo banks of annual killifish. Embryos that enter into Diapause II are capable of surviving for over 600 days (Fig. 5) or the equivalent of two dry seasons. Furthermore, it is probable that had embryos entered Diapause I they would be capable of surviving even longer (Watters 2009). It is unknown whether there are certain years in which pool duration is so short that no fish are able to reach maturity, which would make a multi-year egg bank, and these very long development periods, a necessity. The Pafuri weather station, which is 94 kilometers from our *N. furzeri* collection locality, recorded a mean annual precipitation of 423 mm, with a low of 171 mm in 1947 and a high of 865 mm in 1932 (Fig. S1, Table S1). Without more data linking rainfall levels to pool duration over multiple years it is difficult to come to any firm conclusions regarding (1) the probability of multiple pool fillings per season, and (2) the probability that all pool-fillings within a given season are of such short duration that no fish are able to reach maturity prior to pool dry down. However, there have been several field observations of annual killifish that are consistent with a multi-year embryo bank (discussed in Wourms 1972c; Watters 2009). Pienaar (1968) observed the annual filling, drying, and occasional wash-out flooding of pools containing *Nothobranchius orthonotus* in Kruger National Park over a number of years and asserted (p. 62): "Not all eggs deposited during the particular dry season hatch during the forthcoming rainy season, and some remain dormant in the mud for another year or more, and only hatch when conditions are favourable."

Conclusion

Adaptive phenotypic plasticity is the ideal solution to environmental variance in that the organism can produce the most appropriate phenotype for the expected future environment, but this strategy can only be profitable when cues are indicative of future environmental conditions (Wong and Ackerly 2005). Yet in the field, environmental cues are often imperfect indicators of future environment or season length (Kaplan and Cooper 1984; Moran 1992; Wong and Ackerly 2005; Donaldson-Matasci et al. 2013), such that a diversified bet-hedging strategy can evolve on top of plasticity as a response to the inherent uncertainty (sometimes termed coin-flipping plasticity). That is, theory suggests that when partially reliable cues are available indicating what the future holds the optimal strategy may be to strike a balance between these two modes of adaptation to environmental variation: predictive plasticity and bet-hedging. This empirical study is one of a growing number that considers bet-hedging and phenotypic plasticity together, not as clear cut alternatives but as ends of a spectrum, with the evolved phenotypic response reflecting a blend of the two.

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DATA ARCHIVING

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. Annual precipitation data (mm of rainfall per month) derived from a weather station 94 kilometers from the *Nothobranchius furzeri* collection locality. Station ID: Pafuri, MZ21PFR0. Elevation: 290 meters. Longitude 31.33, Latitude -22.45.
Table S2. Coarse estimates of relevant habitat and environmental parameters for our *N. furzeri* collection locality.
Table S3. Estimates of relevant life-history and development parameters for *N. furzeri*.
Table S4. Summary of *N. furzeri* embryo response to incubation at different combinations of temperature and light.
Figure S1. Climate data from at or near the *N. furzeri* collection locality.
Figure S2. Geographic distribution of the genus *Nothobranchius*.